

Bishal Roy

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Applied AI/ML Engineer specializing in production **Generative AI** — multi-agent systems, Retrieval-Augmented Generation (RAG), and LLM fine-tuning. Ship citation-grounded, low-latency agentic pipelines (FastAPI, LangChain, BM25/FAISS). Smart India Hackathon AIR-2; two research papers under review.

Technical Skills

GenAI & LLMs: Multi-Agent Systems, RAG (BM25/FAISS/ChromaDB), LangChain, LoRA/QLoRA Fine-tuning, Prompt Engineering, HuggingFace Transformers, Groq

ML / DL: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna

MLOps & Data: FastAPI, Docker, Git, CI/CD, MLflow, W&B, Streamlit; Pandas, NumPy, PySpark (basic) **Languages:** Python, SQL

Professional Experience

Interview Kickstart

Apr 2026 – Present

Associate AI Product Manager Intern

Remote

- Built a GenAI evaluation pipeline scoring live-class recordings/transcripts against ratings via LLM-based scoring, auto-flagging the 15% of sessions below 4.5/5 and routing feedback through human-in-the-loop review — lifting average instructor rating from 4.2 to 4.6 in 8 weeks.
- Automated the weekly instructor-ops workflow in Python across 40+ instructors, cutting manual scheduling effort ~87% (~6 hrs/week).

CloudCredits

Jul 2025 – Oct 2025

AI Engineer Intern

Remote

- Built and deployed financial risk models (XGBoost, Random Forest) on 5M+ transaction records at 89% accuracy, cutting false positives 34% in production; engineered a feature pipeline over 3.2M records (+27% precision).

AICTE

Nov 2024 – Dec 2024

AI Engineer Intern

Remote

- Fine-tuned a 7B-parameter open-source LLM (LLaMA-2) with LoRA adapters on 50K Q&A pairs, improving accuracy 72%→91% at sub-2s latency; delivered 8 GenAI/RAG workshops to 600+ faculty across 15 institutions.

Technical Projects

C-TRUST: Clinical Trial Risk Intelligence

Novartis NEST 2.0 — National Semifinalist (Top 30)

Python, FastAPI, Multi-Agent AI

- Architected a 7-agent system with weighted consensus voting (Safety Agent 3.0x) analyzing 23 concurrent studies at ~300ms latency; added a Guardian meta-agent, multi-layer caching, and 331 property-based tests (Hypothesis); deployed via FastAPI + OpenAPI.

Kakehashi: Immigration-Guidance Agent System

Far Away Hackathon (International)

Python, FastAPI, BM25 RAG, Groq, Next.js

- Built a 6-agent system reasoning over official SSW/MOFA sources with citation-backed answers — 51% accuracy vs 4% ungrounded baseline, 0 hallucinations (vs 69) across 22 facts; BM25 RAG, Fernet AES-128 PII encryption, SSE streaming, 22-test CI, EN/HI/JA.

VayuNetra: Urban Air Quality Intelligence

ET AI Hackathon 2026

Python, ML, Geospatial, Groq LLM

- Built an air-quality platform over 900+ stations across 6 cities with 24/48/72h PM2.5 forecasts (HistGradientBoosting) beating persistence baselines 27–50% on RMSE; multi-signal source attribution + LLM multilingual advisories, 14 passing tests.

Achievements & Research

- Smart India Hackathon 2024** — **All India Rank 2** among 49,000+ teams: Mudra, AI Indian Sign Language platform (TensorFlow + MediaPipe), Ministry of Justice.
- Research (Under Review):** First author, JAIR (Problematic Internet Use — QWK optimization & multi-modal features); Co-author, Pattern Recognition, Elsevier (Real-Time ISL Translation).
- BirdCLEF 2026 (Kaggle)** — **Top 10% globally:** 234-species bioacoustic classification (Google Perch v2 + Bayesian fusion, macro ROC-AUC 0.910).

Education

Dr. D. Y. Patil Institute of Technology, Pune

Nov 2022 – May 2026

B.E. in Artificial Intelligence & Data Science — CGPA: 8.99/10 (9.55/10 final semester)